Quality Control Checklist for Survey Instruments

Fillable PDF. Click a blue box to type; tick the small squares. Save with **File › Save** and attach the saved PDF to your assignment. Works best in the free Adobe Acrobat Reader; macOS Preview and web browsers also work. If an answer does not fit its box, continue it on an extra page. Source: `Step_06_Instrument_Construction/quality_control/qc_checklist.md` (rebuild with `tools/make_fillable_pdfs.py`).

Step 6 — Instrument Construction

This checklist should be completed in three passes: **(1)** after drafting the instrument, **(2)** after cognitive interviews, and **(3)** after the pilot study. Retain dated copies of all three passes in this folder.

Stage 1 — Pre-Cognitive-Interview Review (Internal)

1A — Matrix completeness

- [ ] Every item has an entry in the Survey Design Matrix
- [ ] Every item maps to ≥ 1 research question
- [ ] Every item has a documented scale source OR construction rationale
- [ ] Items with no analytic justification have been removed or moved to a "cut list"
- [ ] The variable summary table lists every IV, DV, covariate, and random effect
- [ ] The respondent burden audit is complete and total time is within target range

1B — Item wording

- [ ] No double-barrelled items (asking two things at once)
- [ ] No leading questions (implying a preferred answer)
- [ ] No loaded language (emotionally charged phrasing)
- [ ] Technical jargon is defined or avoided
- [ ] Negatively worded items are flagged (for reverse-scoring in analysis)
- [ ] Reading level is appropriate for target population (aim for ≤ Grade 8 for general public)
- [ ] Items have been translated and back-translated if multilingual delivery is planned
- [ ] For abstract attitudinal constructs (e.g., trust, legitimacy, institutional confidence), consider whether a closed-form item actually captures what respondents mean — Landesvatter & Bauer (2025) used open-ended probing plus supervised machine learning to show that standard trust survey items can miss substantial heterogeneity in how respondents interpret the question; where feasible, pair a closed item with an open-ended probe during cognitive interviewing (Stage 2 below) to check this.

1C — Response options

- [ ] All Likert scales have clearly labelled endpoints and a defined number of points (5 or 7 recommended)
- [ ] All response options are exhaustive (include "other" or "not applicable" where needed)
- [ ] All response options are mutually exclusive
- [ ] Sensitive demographic items include "prefer not to say" and inclusive categories
- [ ] No "double-negative" options (e.g., "Not unfamiliar")
- [ ] Skip logic / branching is documented and tested

1D — Order and structure

- [ ] Demographics are placed at the **end** of the survey
- [ ] Open-ended items appear **before** Likert items if measuring the same construct
- [ ] Within-block randomization is planned where appropriate

- [ ] Randomization plan is documented (for pre-registration)
- [ ] Survey begins with engaging, low-sensitivity items to reduce dropout
- [ ] Attention checks are included (≥ 2 for surveys > 10 min)
- [ ] Attention checks are appropriately framed for the research relationship (community vs. online panel)

1E — Platform / mode checks

- [ ] Survey renders correctly on mobile devices
- [ ] All media files (images, audio, video) load and play correctly
- [ ] Forced-response fields are used only where non-response is truly unacceptable
- [ ] Progress bar is enabled (reduces dropout)
- [ ] Consent page cannot be skipped
- [ ] Response data is stored securely (compliant with IRB / GDPR / PIPEDA as applicable)
- [ ] Participant ID assignment is automated and anonymized

Stage 2 — Cognitive Interview Log

Conduct cognitive interviews with **5–10 participants** from the target population (not colleagues). Use think-aloud protocol: ask participants to verbalize their thought process as they answer each item.

Cognitive interview protocol

1. Introduce the session: "*I'm going to ask you to fill out this survey and talk out loud as you do. There are no right or wrong answers — I want to understand how you interpret each question.*"
2. For each item, probe:
   - "*What does this question mean to you?*"
   - "*How did you decide on that answer?*"
   - "*Was anything confusing or unclear?*"
   - "*Did any of the answer options not fit your situation?*"
3. Record issues in the log below.

Cognitive interview issue log

Item ID	Participant #	Issue type	Verbatim quote (if available)	Resolution
		☐ Comprehension ☐ Recall ☐ Judgment ☐ Response mapping ☐ Sensitive / uncomfortable		☐ Reworded ☐ Response options changed ☐ Item cut ☐ No change — documented

Need more rows? Add a continuation page or attach a spreadsheet with the same columns.

Post-cognitive-interview sign-off

- [ ] All comprehension failures addressed
- [ ] All response-mapping failures addressed
- [ ] Sensitive items reviewed with community liaison (if applicable)
- [ ] Revised instrument saved as `v0.3_post_cognitive_interview_YYYY-MM-DD`

Stage 3 — Pilot Study Protocol

Run a pilot with **30–50 participants** from the target population (or as many as power analysis allows) before full data collection. The pilot tests: comprehension at scale, timing, dropout, data quality, and preliminary psychometric properties.

3A — Pilot data collection checks

- [ ] Pilot sample matches target population demographics
- [ ] Pilot was conducted in the same mode as the main study
- [ ] Consent was re-obtained for pilot participants (they cannot participate again in the main study)
- [ ] Pilot data is stored separately and clearly labelled

3B — Timing and dropout

- [ ] Median completion time matches estimated burden (within ±25%)
- [ ] Page-level drop-off rates have been reviewed
- [ ] Items / pages with high drop-off have been revised or removed

3C — Attention check performance

- [ ] Proportion passing all attention checks: _____ % (flag if < 80%)
- [ ] Speeders (completion time < [expected × 0.4]): _____ % (flag if > 5%)
- [ ] Straight-liners (same response to ≥ 90% of Likert items): _____ % (flag if > 3%)
- [ ] Decision rule for exclusion is documented and pre-registered

3D — Data quality flags

Flag type	Threshold	Observed rate	Action
Failed ≥ 1 attention check	Flag for review	_____ %	
Completion time < threshold	Flag/exclude	_____ %	
Straight-lining (Likert)	Flag for review	_____ %	
Item non-response rate > 10%	Review item	List items:	
Open-ended = gibberish / copy-paste	Exclude	_____ %	

3E — Preliminary psychometrics (for scale items)

- [ ] Internal consistency (Cronbach's α or McDonald's ω) computed for each subscale
- [ ] Acceptable threshold: α / ω ≥ 0.70 for exploratory; ≥ 0.80 for confirmatory
- [ ] Items with item-total correlation < 0.30 flagged for review
- [ ] Confirmatory factor analysis (CFA) fit indices reviewed (if using established scale):
  - CFI ≥ 0.95
  - RMSEA ≤ 0.06
  - SRMR ≤ 0.08
- [ ] Any items revised based on psychometric pilot results are documented

3F — Open-ended content check

- [ ] Random sample of 20% of open-ended responses reviewed for:
  - Relevance to the question
  - Range of responses (not all identical)
  - Absence of offensive / harmful content
  - Absence of identifying information (PII)
- [ ] Thematic coding scheme drafted from pilot responses

3G — Post-pilot sign-off

- [ ] All Stage 3 issues documented with resolutions
- [ ] Final instrument saved as `v1.0_final_pre-registration_YYYY-MM-DD`
- [ ] Instrument version locked and linked to pre-registration
- [ ] Data dictionary completed and archived

Stage 4 — During Data Collection (Ongoing QC)

- [ ] Weekly spot-checks of incoming data for data quality flags
- [ ] Alert thresholds set in survey platform (e.g., flag if completion time drops sharply)
- [ ] Community liaison or field enumerators trained on flagging unusual respondent behaviour
- [ ] Any mid-study changes to instrument are documented with date, rationale, and effect on data comparability

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Stage 5 — Post-Collection Data Cleaning Checklist

- [ ] Exclusion criteria applied consistently per pre-registered decision rule
- [ ] All exclusions documented in a separate log with reason codes
- [ ] Missing data pattern examined (MCAR / MAR / MNAR assessment)
- [ ] Reverse-scored items recoded before analysis
- [ ] Composite scale scores computed per pre-registered formula
- [ ] Data dictionary shared with all analysts
- [ ] Raw data and cleaned data stored separately (raw data never overwritten)
- [ ] Data cleaning script version-controlled (Git)

References

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